

JobAlign

AI-Powered Job & Skill Alignment Platform

Requirement Analysis Document

Software Requirements Specification (SRS)

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This document is the controlled requirements baseline for the JobAlign project. Any change to the requirements recorded here must follow the change control process defined in Section 1.5.

Document Control

Revision History

Version	Date	Author	Summary of Change
0.1	05 Aug 2026	Business Analyst	Initial draft — problem statement, scope and stakeholders.
0.2	09 Aug 2026	Business Analyst	Functional requirements added for capture, extraction and matching.
0.3	11 Aug 2026	Business Analyst	Resume analysis, normalization, duplicate detection and roadmap modules included.
1.0	12 Aug 2026	Business Analyst	Baselined for review. Non-functional requirements, risks and traceability completed.

Distribution List

Role	Purpose of Distribution
Project Sponsor	Approval of scope and business objectives.
Technical Lead	Input to system design and database design.
Development Team	Basis for sprint planning and development tasks.
Quality Assurance	Basis for test case design and acceptance testing.

Approval

Name	Role	Signature	Date
	Project Sponsor		
	Technical Lead		
	Quality Assurance Lead		

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1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for JobAlign, an application that converts unstructured job postings into structured, comparable data and measures how well a candidate's skills align with each opportunity. It establishes the agreed requirements baseline before system design, database design and development begin.

1.2 Intended Audience

This document is intended for the project sponsor, the technical lead, the development team and the quality assurance team. Sections 2 to 4 describe the business need and are relevant to all readers. Sections 5 to 10 provide the detail required for design, development and testing.

1.3 Definitions and Abbreviations

Term	Definition
Candidate	A registered user who saves job postings and maintains a profile.
Job Posting	The original, unmodified text of an advertised role as captured by the system.
Normalization	Conversion of varying expressions of the same value into one standard form.
Master Skill	The single approved name for a skill, to which all variant names resolve.
Match Score	A calculated measure of how closely a candidate profile fits a job posting.
Skill Gap	The set of skills required by a posting that are absent from the candidate profile.
Extraction	The process of deriving structured fields from unstructured posting or resume text.
SRS	Software Requirements Specification.

1.4 Document Conventions

Each requirement carries a unique identifier. Functional requirements are numbered FR-nn, business rules BR-nn, and non-functional requirements NFR-nn. Priority is recorded as **M** (Must have — required for first release), **S** (Should have — important but deferrable) and **C** (Could have — included if capacity allows). The word "shall" indicates a mandatory requirement.

1.5 Change Control

Once baselined, any change to a requirement in this document must be raised as a change request, assessed for impact on scope, schedule and effort, and approved by the project sponsor. Approved changes are recorded in the revision history and the document version is incremented.

2. Business Context

2.1 Problem Statement

Candidates collect opportunities from multiple portals, referrals and message groups. Job postings are written in inconsistent, marketing-oriented language, and the details that most influence a decision — required experience, salary, location, work mode and expected skills — are buried in long paragraphs, expressed in incompatible formats, or omitted entirely.

The same skill appears under several names, salaries are quoted per year, per month or per hour, and locations are written differently by every employer. As a result candidates record details by hand, cannot compare two openings on equal terms, and have no dependable view of which skills they are missing. Applications are submitted untargeted and skill development is based on guesswork rather than evidence.

2.2 Business Objectives

Ref	Objective
BO-01	Provide a single system of record for the job postings a candidate is considering.
BO-02	Convert unstructured posting text into structured, consistent and comparable data.
BO-03	Standardize skills, salaries and locations so that postings can be compared on equal terms.
BO-04	Measure the fit between a candidate and a posting and express it as an understandable score.
BO-05	Identify skill gaps and convert them into a prioritized learning plan.
BO-06	Reduce the manual effort of recording, comparing and tracking opportunities.

2.3 Success Measures

Measure	Target
Time to record and structure one posting	Under one minute, compared with manual entry today.
Extraction quality	No more than two fields corrected by the candidate in 80% of postings.
Skill resolution	At least 90% of extracted skills resolve to an existing master skill.
Adoption	A candidate saves five or more postings within the first week of use.

3. Scope

In Scope	Out of Scope (Version 1.0)
Capture of job postings from pasted text or a link	Applying to jobs from within the application
AI extraction of posting details into structured fields	Scheduled crawling of external job portals
Normalization of skills, salaries and locations	Employer accounts and the publishing of vacancies
Duplicate detection and posting quality assessment	Resume generation or automated resume rewriting
Candidate profile and resume-based skill extraction	Interview scheduling and in-app messaging
Match scoring, skill gap analysis and learning roadmap	Payment, subscription and billing
Job comparison, AI summaries and candidate dashboard	Mobile native applications
Authentication, role-based access and administration	Integration with external learning platforms

4. Stakeholders and Users

4.1 Stakeholders

Stakeholder	Interest / Need
Candidate	Save and compare opportunities, understand fit, and know which skills to build next.

Stakeholder	Interest / Need
Administrator	Maintain users, the master skill list and system configuration.
Project Sponsor	Delivery of the agreed scope within schedule and budget.
Development Team	Clear, testable requirements to design and build against.
Quality Assurance	Acceptance criteria that can be converted into test cases.

4.2 Actors

Actor	Type	Responsibility in the System
Candidate	Primary	Owens postings, resume and profile; captures, reviews, corrects, compares and tracks.
Administrator	Primary	Manages users, roles, the master skill list and system settings.
AI Service	Supporting	Converts unstructured posting and resume text into structured fields on request.

The AI service and the database are components of the system infrastructure and are not user actors. The application depends on the AI service but shall continue to operate in a reduced form when that service is unavailable (see FR-19 and NFR-05).

4.3 Role and Permission Matrix

Function	Candidate	Admin
Capture and manage own job postings	Yes	No
View and correct extracted details of own postings	Yes	No
Maintain own profile and upload resume	Yes	No
View own match scores, skill gaps and roadmap	Yes	No
Manage users and roles	No	Yes
Maintain master skill list and aliases	No	Yes
Configure extraction settings	No	Yes
View system-wide reports	No	Yes

5. System Overview

The system is organised into nine functional modules. Each module groups a set of related requirements and forms a candidate unit for sprint planning.

Ref	Module	Responsibility
M-01	Authentication and Access	Registration, sign-in and role-based access control.
M-02	Job Posting Capture	Recording postings and preserving their original text.
M-03	Extraction and Normalization	Structuring posting text and standardizing skills, salary and location.
M-04	Quality and Duplicates	Detecting incomplete postings and repeated entries.
M-05	Profile and Resume	Candidate profile and resume-based skill extraction.

Ref	Module	Responsibility
M-06	Matching and Scoring	Comparing profile against posting and calculating scores.
M-07	Gap, Feedback and Roadmap	Identifying gaps and prioritizing what to learn.
M-08	Comparison and Dashboard	Summaries, side-by-side comparison and progress views.
M-09	Administration	User management, master data and reporting.

6. Functional Requirements

6.1 Authentication and Access Control (M-01)

ID	Requirement	Pri
FR-01	The system shall allow a user to register with a unique email address and a password.	M
FR-02	The system shall authenticate a user before granting access to any candidate or administrative function.	M
FR-03	The system shall assign every user the role of Candidate or Administrator and enforce access according to the matrix in Section 4.3.	M
FR-04	The system shall allow a user to sign out and shall end the session after a defined period of inactivity.	M
FR-05	The system shall allow a user to reset a forgotten password through a verified email link.	S

6.2 Job Posting Capture (M-02)

ID	Requirement	Pri
FR-06	The system shall allow a candidate to save a job posting by pasting its text.	M
FR-07	The system shall allow a candidate to save a job posting by providing a link, and shall retrieve the posting text from that link.	S
FR-08	The system shall store the original posting text without modification and retain it for the life of the posting.	M
FR-09	The system shall generate a unique reference for each saved posting and set its initial status to New.	M
FR-10	The system shall allow a candidate to record the source of a posting and the date it was captured.	S
FR-11	The system shall allow a candidate to archive or delete a saved posting.	S

6.3 AI Extraction and Normalization (M-03)

ID	Requirement	Pri
FR-12	The system shall extract the job title, company name, location, work mode, required experience, salary and responsibilities from the posting text.	M
FR-13	The system shall extract the skills stated in a posting and shall classify each as required or preferred.	M
FR-14	The system shall resolve skill name variations to a single master skill, so that entries such as "C Sharp", "C-Sharp" and "C#" are treated as one skill.	M
FR-15	The system shall normalize salary values expressed in different currencies, periods and formats into a common comparable form, retaining the originally stated value.	M

ID	Requirement	Pri
FR-16	The system shall normalize location names so that variations of the same place resolve to a single location entry.	S
FR-17	The system shall record any detail not stated in the posting as "Not specified" and shall not substitute a default or estimated value.	M
FR-18	The system shall present all extracted details to the candidate for review and shall allow any detail to be corrected before the posting is confirmed.	M
FR-19	Where extraction fails or the AI service is unavailable, the system shall retain the posting with the status Pending and shall offer both a retry and manual entry.	M
FR-20	The system shall record a confidence indicator against each extracted detail and shall flag low-confidence details for review.	S
FR-21	The system shall allow extraction to be re-run against the stored original text without re-capturing the posting.	S

6.4 Job Quality and Duplicate Detection (M-04)

ID	Requirement	Pri
FR-22	The system shall assess each posting for completeness and shall report which expected details are absent.	S
FR-23	The system shall present a posting quality indicator derived from the completeness assessment.	C
FR-24	The system shall identify postings already saved by the candidate that are identical or substantially similar to a newly captured posting.	S
FR-25	The system shall warn the candidate when a suspected duplicate is detected and shall allow the candidate to keep both entries or discard the new one.	S
FR-26	The system shall allow a candidate to link two postings as the same role advertised through different sources.	C

6.5 Candidate Profile and Resume Analysis (M-05)

ID	Requirement	Pri
FR-27	The system shall allow a candidate to maintain a profile containing personal details, education, work experience, projects and certifications.	M
FR-28	The system shall allow a candidate to add skills to the profile and to record a proficiency level for each skill.	M
FR-29	The system shall resolve profile skills to the master skill list using the same rules applied to posting skills.	M
FR-30	The system shall allow a candidate to upload a resume in a supported document format.	S
FR-31	The system shall extract skills, education and work experience from an uploaded resume.	S
FR-32	The system shall present skills extracted from a resume as suggestions, and shall add them to the profile only after the candidate confirms them.	S
FR-33	The system shall calculate the candidate's total years of experience from the profile for use in scoring.	M
FR-34	The system shall allow a candidate to remove a skill or an uploaded resume from the profile.	M

6.6 Job Matching and Match Score (M-06)

ID	Requirement	Pri
FR-35	The system shall compare the skills held in the candidate profile against the skills required by each confirmed posting.	M
FR-36	The system shall calculate a required-skill score representing the proportion of required skills the candidate holds.	M
FR-37	The system shall calculate a preferred-skill score representing the proportion of preferred skills the candidate holds.	M
FR-38	The system shall calculate an experience score by comparing the candidate's total experience against the experience required by the posting.	M
FR-39	The system shall calculate an overall match score from the required-skill, preferred-skill and experience scores using defined weightings.	M
FR-40	The system shall display the component scores alongside the overall score so that the result can be explained.	S
FR-41	The system shall recalculate match scores for all confirmed postings whenever the candidate profile changes.	M

6.7 Skill Gap, Feedback and Learning Roadmap (M-07)

ID	Requirement	Pri
FR-42	The system shall display, for each posting, the skills the candidate holds and the skills the candidate is missing.	M
FR-43	The system shall distinguish missing required skills from missing preferred skills.	M
FR-44	The system shall generate written feedback for a posting describing the candidate's strengths and weaknesses in relation to that role.	S
FR-45	The system shall identify the skills most frequently missing across all of the candidate's saved postings.	S
FR-46	The system shall produce a learning roadmap that orders missing skills by how often they are required and whether they are required or preferred.	S
FR-47	The system shall allow a candidate to mark a roadmap skill as in progress or completed, and shall reflect completion in the profile once confirmed.	C

6.8 Comparison, Summary and Dashboard (M-08)

ID	Requirement	Pri
FR-48	The system shall generate a short summary of a lengthy job description.	S
FR-49	The system shall allow a candidate to compare selected postings side by side on skills, salary, experience, location and match score.	S
FR-50	The system shall allow a candidate to filter saved postings by work mode, location, experience range and match score.	M
FR-51	The system shall allow a candidate to sort saved postings by match score, salary or date captured.	M
FR-52	The system shall provide a dashboard presenting match scores, current skill gaps, the number of postings analysed and progress against the learning roadmap.	M

ID	Requirement	Pri
FR-53	The system shall allow a candidate to record the application status of a posting as Saved, Applied, Interview, Rejected or Closed.	S
FR-54	The system shall exclude postings with the status Pending from comparison and from dashboard scoring figures.	M

6.9 Administration (M-09)

ID	Requirement	Pri
FR-55	The system shall allow an administrator to create, deactivate and reactivate user accounts.	M
FR-56	The system shall allow an administrator to assign and change a user's role.	M
FR-57	The system shall allow an administrator to add, edit and deactivate entries in the master skill list.	M
FR-58	The system shall allow an administrator to maintain skill aliases and to merge two master skills into one.	M
FR-59	The system shall allow an administrator to view a report of extraction volumes, failure rates and the frequency of candidate corrections.	C
FR-60	The system shall record an audit entry for administrative changes to users, roles and master data.	S

7. Business Rules

ID	Rule
BR-01	The original posting text is never altered. All extracted and normalized values are derived data and may be regenerated at any time.
BR-02	A detail not stated in a posting is recorded as "Not specified". It is never recorded as zero, blank or an estimated value.
BR-03	A correction made by the candidate takes precedence over the extracted value and is preserved when extraction is re-run.
BR-04	Every skill, whether taken from a posting, a resume or a profile, resolves to exactly one entry in the master skill list.
BR-05	A normalized salary always retains the originally stated value and period alongside the comparable value.
BR-06	Match scores are calculated only from skills confirmed by the candidate. Skills suggested from a resume do not affect scoring until confirmed.
BR-07	Required skills carry a greater weight in the overall match score than preferred skills.
BR-08	A posting with the status Pending is excluded from scoring, comparison and dashboard figures until its details are confirmed.
BR-09	A posting, resume and profile are visible only to the candidate who owns them. Administrators may manage accounts but may not view candidate postings or resumes.
BR-10	Where a posting states no salary, no comparable salary value is produced and the posting is excluded from salary-based sorting.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Extraction of a posting shall complete within 10 seconds for 95% of submissions.
NFR-02	Performance	The saved-postings list and the dashboard shall render within 2 seconds for up to 500 postings.
NFR-03	Performance	Match scores for all postings shall be recalculated within 30 seconds of a profile change.
NFR-04	Security	All functions shall require authentication, and access shall be enforced by role on the server side.
NFR-05	Security	Passwords shall be stored using a salted one-way hash and all traffic shall be transmitted over HTTPS.
NFR-06	Reliability	Unavailability of the AI service shall not prevent a posting or resume from being saved.
NFR-07	Accuracy	Extracted values shall never be invented; unstated details shall be reported as unspecified.
NFR-08	Auditability	The original text and the extraction configuration version shall be retained so that any result can be reproduced and explained.
NFR-09	Privacy	Only the content required for extraction shall be transmitted to the AI service, and a candidate shall be able to delete their resume and profile data.
NFR-10	Usability	The interface shall be responsive and operable on desktop, tablet and mobile screen widths.
NFR-11	Maintainability	Extraction logic shall be isolated behind a service interface so that the AI provider can be replaced without changes to the application.
NFR-12	Scalability	The system shall support growth in stored postings and concurrent users without architectural redesign.
NFR-13	Cost	Extraction shall be performed once per posting and the result stored, so that repeated viewing incurs no further processing cost.

9. External Interface and Technology Requirements

9.1 User Interface

The application shall present a browser-based interface using a responsive layout. Extracted details shall always be shown in a reviewable form before they are committed, and any value that was normalized shall display the originally stated value alongside it.

9.2 AI Service Interface

The application shall communicate with an external AI service over HTTPS to perform extraction, normalization, summarization and feedback generation. Requests and responses shall use a structured data format. The application shall handle service errors, timeouts and rate limits without loss of candidate data, in accordance with FR-19 and NFR-06.

9.3 Technology Constraints

Layer	Technology
Programming Language	C#

Layer	Technology
Web Application	ASP.NET Core MVC
Service Layer	ASP.NET Core Web API
Data Access	Entity Framework Core
Database	SQL Server
User Interface	Bootstrap
Extraction and Analysis	External AI API

10. Data Requirements

The following information shall be held by the system. Entity structure, keys and relationships will be defined during database design.

Information	Details Held
User	Identity, credentials, role and account status.
Candidate Profile	Personal details, education, work experience, projects, certifications and total experience.
Resume	Uploaded document, upload date and extraction status.
Job Posting	Source, original text, reference, capture date, application status and owning candidate.
Extracted Detail	Title, company, location, work mode, experience range, salary, responsibilities, confidence and configuration version.
Master Skill	Approved skill name, category and active status.
Skill Alias	Alternative name and the master skill it resolves to.
Posting Skill	Skill linked to a posting, classified as required or preferred.
Profile Skill	Skill linked to a candidate, with proficiency level and source of confirmation.
Match Result	Overall score, required-skill score, preferred-skill score, experience score and calculation date.
Skill Gap	Missing skills for a posting, classified as required or preferred.
Roadmap Item	Skill, priority, status and date last updated.
Audit Entry	Action performed, user, affected record and timestamp.

11. User Stories

Actor	User Story
Candidate	As a candidate, I want to paste a job posting so that its key details are structured for me instead of being recorded by hand.
Candidate	As a candidate, I want incorrectly extracted details to be correctable so that my saved data remains accurate.
Candidate	As a candidate, I want salaries and locations expressed consistently so that I can compare postings on equal terms.
Candidate	As a candidate, I want to upload my resume so that my skills are captured without entering them one by one.

Actor	User Story
Candidate	As a candidate, I want to see a match score with its component parts so that I understand why a role fits or does not.
Candidate	As a candidate, I want to see which required skills I am missing so that I know what to learn before applying.
Candidate	As a candidate, I want a prioritized learning roadmap so that I spend my time on the skills demanded most often.
Candidate	As a candidate, I want to be warned about duplicate postings so that my saved list stays clean.
Administrator	As an administrator, I want to merge duplicate skills so that the same skill is not held under several names.

12. Acceptance Criteria

User Story: As a candidate, I want to paste a job posting so that its key details are structured and saved for comparison. (FR-06, FR-08, FR-09, FR-12 to FR-19)

#	Acceptance Criterion
1	The candidate is logged in.
2	The posting text contains at least 100 characters, otherwise a validation message is displayed.
3	The posting is saved with its original text unchanged.
4	A unique reference is generated and the status is set to New.
5	The title, company, location, work mode, experience, salary and skills are extracted.
6	Extracted skills are resolved to master skills and classified as required or preferred.
7	Salary and location values are normalized, with the originally stated values retained.
8	Any detail not stated in the posting is recorded as "Not specified".
9	All extracted details are presented for review and any detail can be corrected before confirmation.
10	On confirmation, the status changes to Confirmed and a match score is calculated.
11	If extraction fails, the posting remains saved with the status Pending and a retry is offered.

13. Assumptions, Dependencies and Constraints

Assumptions

- Candidates have the right to store, for personal use, the postings they capture.
- Posting and resume content is supplied in English.
- A candidate maintains an accurate profile; match results are only as reliable as the profile behind them.
- A master skill list is available at go-live and is maintained thereafter by an administrator.

Dependencies

- Availability and continued support of the external AI service used for extraction and analysis.
- Availability of a SQL Server instance and a hosting environment for the application.

Constraints

- Each extraction carries a processing cost, so extraction is performed once per posting and the result is stored.
- Extraction quality depends on the quality of the source text; a vague posting yields fewer confirmed details.
- Some job portals restrict automated access, so link-based capture may not succeed for every source.
- The application must be delivered using the technology stack recorded in Section 9.3.

14. Risks

Ref	Risk	Impact	Mitigation
R-01	Extraction returns inaccurate details, reducing trust in match scores.	High	Mandatory candidate review before confirmation (FR-18); confidence flagging (FR-20).
R-02	AI service outage or rate limiting interrupts capture.	High	Postings saved independently of extraction; Pending status with retry and manual entry (FR-19).
R-03	Skill list fragments through unresolved name variants.	Medium	Alias resolution (FR-14) and administrative merge capability (FR-58).
R-04	Extraction cost grows faster than expected.	Medium	One extraction per posting with stored results (NFR-13); re-run only on request.
R-05	Resume content introduces personal data handling obligations.	Medium	Minimal transmission to the AI service and candidate-initiated deletion (NFR-09).
R-06	Scope expands during development through added AI features.	Medium	Priority classification in Section 6 and change control in Section 1.5.

15. Requirements Traceability Matrix

Objective	Business Objective	Module	Requirements
BO-01	Single system of record for postings	M-02	FR-06 to FR-11
BO-02	Structured data from unstructured text	M-03	FR-12, FR-13, FR-17 to FR-21
BO-03	Comparable skills, salaries and locations	M-03, M-04	FR-14, FR-15, FR-16, FR-22 to FR-26
BO-04	Measurable fit between candidate and role	M-05, M-06	FR-27 to FR-41
BO-05	Skill gaps converted to a learning plan	M-07	FR-42 to FR-47
BO-06	Reduced manual effort in tracking opportunities	M-08	FR-48 to FR-54
—	Controlled access and master data quality	M-01, M-09	FR-01 to FR-05, FR-55 to FR-60

End of document. This requirements baseline is the input to the System Design, Database Design and Sprint Planning stages. Requirements marked **M** constitute the minimum viable release.